Phase 1 key acquisition guide

NetAssure Protocol Guardian (NAPG) — issued 2026-05-17

This guide walks through obtaining every external service credential needed for Phase 1 of NAPG. Each key has a free or starter tier sufficient for testnet work. Paste each value into your local `.env` file (copy `.env.example` to `.env` first).

Total estimated time: 30 to 45 minutes, assuming you have a Coinbase account already.

Checklist

- [ ] Copy `.env.example` to `.env` at the repo root
- [ ] Anthropic API key created and pasted
- [ ] CDP project created, Secret API key downloaded, name and private key pasted
- [ ] CDP wallet ID left blank (AgentKit creates it on first run)
- [ ] Base Sepolia RPC URL pasted (Alchemy recommended)
- [ ] 0x API key pasted
- [ ] Base Sepolia wallet funded with dust via faucet (after first run creates the wallet)

1. ANTHROPIC_API_KEY — reasoning layer

What it powers: sub-phase 1C reasoning layer. The agent sends protocol-state JSON to Claude and receives a structured decision.

1. Sign in at console.anthropic.com
2. Add billing credits (minimum 5 USD — required even for trial access)
3. Settings, then API Keys, then Create Key. Name it `napg-dev`.
4. Copy the key once (it is not shown again) and paste into `.env`:

```
ANTHROPIC_API_KEY=sk-ant-api03-...
```

> **Gotcha:** this is a separate billing surface from your Claude Code subscription. We will lean heavily on prompt caching to keep costs low.

2. CDP wallet keys — Coinbase Developer Platform

What they power: sub-phase 1C MPC wallet creation, transaction signing, and the `move_to_safe_contract` action via CDP AgentKit.

1. Sign in at portal.cdp.coinbase.com with your Coinbase account
2. Create a project (e.g. `napg`)
3. API Keys, then Create API key. Choose **Secret API Key** — not the read-only or Sign-In-With-Coinbase variants

4. Coinbase shows the key once and offers a JSON download. Save the JSON file in a secure password manager.
5. From the JSON, paste the `name` and `privateKey` fields into `.env`:

```
CDP_API_KEY_NAME=organizations/abc.../apiKeys/123...
CDP_API_KEY_PRIVATE_KEY="-----BEGIN EC PRIVATE KEY-----\n...\n-----END EC PRIVATE KEY-----
\n"
CDP_WALLET_ID=
```

Leave `CDP_WALLET_ID` blank. AgentKit creates an MPC wallet on first run and the build step will persist the ID at that point.

Faucet (after first run)

Once the wallet is created, fund it with a dust amount of Base Sepolia ETH:

- coinbase.com/faucets/base-ethereum-sepolia-faucet
- Backup: alchemy.com/faucets/base-sepolia

> **Policy:** PLAN.md caps the Phase 1 wallet to a known dust amount. Do not over-fund.

3. BASE_SEPOLIA_RPC_URL — Web3.py heartbeat

What it powers: sub-phase 1B RPC heartbeat (latency, block height, reorg detection) and 1E end-to-end migration.

Provider	Free tier	How to get
Alchemy *(recommended)*	Generous; reliable	alchemy.com -> Sign up -> Create App -> Chain: Base, Network: Base Sepolia -> copy HTTPS URL
Infura	Yes	infura.io -> similar flow
Public	Rate limited	Use `https://sepolia.base.org` without an account; fine for early validation only

```
BASE_SEPOLIA_RPC_URL=https://base-sepolia.g.alchemy.com/v2/YOUR_KEY
```

> **Security:** Phase 0 baseline requires HTTPS only. All three providers above qualify; the agent rejects `http://` at config-load time.

4. ZEROX_API_KEY — 0x liquidity probe

What it powers: sub-phase 1B Aerodrome USDC/USDbC pool liquidity probe.

1. Sign in at dashboard.0x.org
2. Create an API key. The free tier is rate-limited but enough for Phase 1 polling.
3. Paste into `.env`:

```
ZEROX_API_KEY=...
```

Not needed for Phase 1

- `BASE_RPC_URL` (mainnet) — Phase 5 cutover
- `TENDERLY_ACCOUNT_SLUG` / `TENDERLY_PROJECT_SLUG` / `TENDERLY_ACCESS_KEY` — Phase 2 (AgentShield simulation)
- `NEYNAR_API_KEY`, `FARCASTER_SIGNER_UUID` — Phase 3 (Sign In With Farcaster and cast distribution)

Verify .env

Once filled in, from the repo root:

```
grep -v '^#\|^$' .env
```

You should see at minimum:

```
ANTHROPIC_API_KEY=sk-ant-...
CDP_API_KEY_NAME=organizations/...
CDP_API_KEY_PRIVATE_KEY=-----BEGIN EC PRIVATE KEY-----...
CDP_WALLET_ID=
BASE_SEPOLIA_RPC_URL=https://...
ZEROX_API_KEY=...
NAPG_NETWORK=base-sepolia
NAPG_AUDIT_DB=./audit.db
```

> **Never commit .env.** It is gitignored at line 44 of `.gitignore`. The gitleaks pre-commit hook is the local backstop; the CI security job is the remote backstop.

Generated for Samuel Apakuki Nayacakalou, NAPG Phase 1 onboarding. See `PLAN.md` Phase 1 and `.env.example` for the canonical key list.